JMLR MLOSS reviewer comments and response status

kodama-cpp release candidate 0.1.0

Internal submission-readiness report

This report separates completed manuscript/repository revisions from evidence that cannot be represented as complete until the remaining benchmark matrix, tagged archive, and author declarations are available.

Comment	Response	Status
KODAMA 2.4.1/2.4 comparison	A matched MetRef KNN benchmark uses KODAMA 2.4.1/2.4 as the sole predecessor performance comparator against kodama-cpp single-core CPU, four-core CPU, and CUDA paths. Later KODAMA releases are not performance baselines.	Completed on chiamaka
Frozen public API	Version 0.1.0 macros, SemVer policy, CMake package version, wrapper versions, and a compile-link API snapshot test are present.	Completed locally
Version/tag/archive	The manuscript records public core commit 0b8b873 and R-wrapper commit 7090592. A release checklist and clean-tag archive script prevent an uncommitted or dirty artifact from being cited; a DOI is optional.	External tag and deposit pending
License provenance	Pinned upstream snapshots, per-component licensing, SPDX tests, retained licenses, and the Metal Apache exception are documented.	Engineering audit complete; coauthor contribution confirmation remains
MLOSS page budget	The detailed manuscript is retained as a technical supplement; a separate official-style main paper targets four description pages plus references.	Completed; four description pages plus references verified
CUDA release build	Commit a32f718 passed all four configured license, numerical/core, public-API, and float32 smoke tests in the CUDA 13.0 build on chiamaka.	Completed on chiamaka
Continuous integration	GitHub Actions builds and tests the CPU core on Linux/macOS, native Metal on macOS, and the R/Python wrappers on Ubuntu. Separate coverage and API-documentation workflows passed for a32f718.	Completed at a32f718
User documentation	A compact C++/R/Python walkthrough and generated Doxygen API are published through GitHub Pages.	Completed and URL verified
Active user community	The cover letter reports dated CRANlogs evidence for the predecessor R package and explicitly discloses that the new library has no public adoption metrics yet.	Independent kodama-cpp usage evidence pending
Local wrapper release checks	R CMD check --as-cran passed with no errors or warnings and one environment-only macOS detritus NOTE; an isolated wheel built against the installed core passed eight tests including physical-Metal visualization diagnostics.	Completed locally on current worktree

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Python publication	The tested Python wrapper is included in the public core repository and exercised by CI.	Standalone public repository pending
Coverage depth	LLVM reports 80.35% line and 68.79% branch CPU coverage. Dedicated graph/clustering contracts and the public binary UMAP compaction/replay contract exercise generic raw/graph orchestration, stochastic invariants, degenerate PLS-LDA, and visualization behavior on portable and physical-Metal builds.	Improved locally; resident-IVF and accelerator branches pending
Large-sample SIMPLS range	An overflow-only power-norm guard restores all 50 requested components on MNIST70k. Three-replicate CPU/Metal validation, a synthetic regression, AddressSanitizer, and physical Metal CTest pass.	Local evidence complete; frozen CUDA confirmation pending
CPU PLS-LDA memory traffic	Training class sums and the latent Gram matrix are streamed without complete score matrices. Four-dataset predictions match; TabulaMuris is 1.47x faster with a 0.587 peak-memory ratio.	Accepted locally
M/Tcycle rationale	Named MetRef/USPS sweeps report CV accuracy, ARI, active classes, runtime, and agreement-graph convergence; M=100 is justified by ensemble precision rather than external-label tuning.	Completed on CUDA
Full-cycle local backend controls	Shared-graph MetRef, USPS, ImageSegmentation, PageBlocks, PenDigits, and SatImage runs use M=Tcycle=100, CPU4 and physical Metal. Raw COIL20 adds paired KNN timing and a censored PLS-LDA stress cell. Landmark timing is exposed separately. SatImage records Metal 1.06x slower for KNN and 5.87x slower for PLS-LDA, with KODAMA silhouettes below classic UMAP.	Completed locally; clean tagged CPU4/CUDA replication pending
Final ablations	The release driver fixes M=100, Tcycle=100, landmarks, k, ncomp, splitting, graph correction, backend, wrappers, and external metrics before inspection.	Eleven-dataset CUDA archive analyzed with three seed-matched replicates; matched KODAMA 2.4.1 predecessor reruns and the frozen confirmatory matrix remain required
Visualization claims	Classic and KODAMA UMAP/openTSNE are paired by dataset, seed, and CUDA profile. Dataset-level seed medians, 100,000-resample bootstrap intervals, exact Wilcoxon and sign tests, and Holm adjustment are reported. The archive is explicitly exploratory and no universal improvement is claimed.	PLS-LDA UMAP: 6/10, median +0.025, 95% CI [-0.067,+0.084], Holm p=0.557; KNN openTSNE: 0/10, Holm p=0.0078
HPC provenance	Uploaded run manifests preserve dataset and parameter metadata but record git_commit=NA. The manuscript does not retrospectively assign them to a source revision.	Clean tagged confirmatory rerun with source checksum and container digest pending

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Cover letter	A track-specific draft includes prior-publication disclosure, license, URL, version, software delta, and explicit author-only fields.	Draft complete; author fields pending
Release sustainability	README, changelog, contribution guide, API policy, prior-version delta, validation protocol, and release checklist are included.	Completed locally